

Frontend Pages to Backend Routes Mapping

This document maps each frontend page to its corresponding backend route(s) and service(s).

Page-to-Route Mapping Table

Frontend Page	Backend Route	Description	Status
Login.tsx	/api/auth	User authentication	☒ Connected
Register.tsx	/api/auth	User registration	☒ Connected
Dashboard.tsx	/api/analytics/dashboard	Main dashboard	☒ Connected
Home.tsx	N/A	Landing page (no API)	☒ N/A
About.tsx	N/A	About page (no API)	☒ N/A
NotFound.tsx	N/A	404 page (no API)	☒ N/A

Admin Pages

Frontend Page	Backend Route	Description	Status
Users.tsx	/api/users	User management	☒ Connected
Orders.tsx	/api/orders	Order management	☒ Connected
Products.tsx	/api/products	Product management	☒ Connected
Inventory.tsx	/api/inventory	Inventory management	☒ Connected
Production.tsx	/api/production	Production tracking	☒ Connected
Settings.tsx	/api/configuration	System settings	☒ Connected

Analytics Pages

Frontend Page	Backend Route	Description	Status
Analytics.tsx	/api/analytics	General analytics	☒ Connected
ProductionAnalytics.tsx	/api/analytics	Production analytics	☒ Connected
InventoryAnalytics.tsx	/api/analytics	Inventory analytics	☒ Connected

System Administration Pages

Frontend Page	Backend Route	Description	Status
SystemLogs.tsx	/api/system-logs	System logs viewer	☒ Connected

Frontend Page	Backend Route	Description	Status
ApiKeys.tsx	/api/api-keys	API key management	☒ Connected
NotificationConfigs.tsx	/api/notification-configs	Notification settings	☒ Connected
SystemConfigurationDashboard.tsx	/api/configuration	System configuration	☒ Connected

Data Management Pages

Frontend Page	Backend Route	Description	Status
DataManagementDashboard.tsx	/api/data-management	Data management	☒ Connected

Quality Control Pages

Frontend Page	Backend Route	Description	Status
QualityControlDashboard.tsx	/api/quality	Quality control	☒ Connected

Operations Pages

Frontend Page	Backend Route	Description	Status
OperationsManagement.tsx	/api/production, /api/harvest	Operations management	☒ Connected
ProcessingDashboard.tsx	/api/harvest	Processing dashboard	☒ Connected
Worksheets.tsx	/api/production	Worksheets management	☒ Connected
PoultryCareDashboard.tsx	/api/production, /api/medication	Poultry care	☒ Connected

Employee Pages

Frontend Page	Backend Route	Description	Status
EmployeeDashboard.tsx	/api/employees	Employee dashboard	☒ Connected
EmployeeProfile.tsx	/api/employees	Employee profile	☒ Connected
EmployeeSchedule.tsx	/api/employees	Employee schedule	☒ Connected
EmployerDashboard.tsx	/api/employees	Employer dashboard	☒ Connected
EmployerEmployees.tsx	/api/employees	Employee management	☒ Connected

Customer Pages

Frontend Page	Backend Route	Description	Status
CustomerDashboard.tsx	/api/crm, /api/orders	Customer dashboard	☒ Connected
CustomerOrders.tsx	/api/orders	Customer orders	☒ Connected
CustomerCRM.tsx	/api/crm	Customer CRM	☒ Connected
Shop.tsx	/api/products, /api/orders	Shopping page	☒ Connected

Restaurant Pages

Frontend Page	Backend Route	Description	Status
RestaurantDashboard.tsx	/api/orders, /api/products	Restaurant dashboard	☒ Connected
RestaurantOrders.tsx	/api/orders	Restaurant orders	☒ Connected
RestaurantInventory.tsx	/api/inventory	Restaurant inventory	☒ Connected

Demo Pages

Frontend Page	Backend Route	Description	Status
ModelBindingDemo.tsx	N/A	Demo page (no API)	☒ N/A

Detailed Mapping by Feature

Authentication & Users

Frontend:		Backend:
- Login.tsx	→	/api/auth/login
- Register.tsx	→	/api/auth/register
- Users.tsx	→	/api/users (CRUD)

Products & Shop

Frontend:		Backend:
- Products.tsx	→	/api/products (CRUD)
- Shop.tsx	→	/api/products, /api/orders

Orders & Payments

Frontend:		Backend:
- Orders.tsx	→	/api/orders (CRUD)
- CustomerOrders.tsx	→	/api/orders (user orders)
- RestaurantOrders.tsx	→	/api/orders (restaurant orders)

Inventory & Production

Frontend:		Backend:
- Inventory.tsx	→	/api/inventory
- RestaurantInventory.tsx	→	/api/inventory
- InventoryAnalytics.tsx	→	/api/analytics, /api/inventory
- Production.tsx	→	/api/production
- ProductionAnalytics.tsx	→	/api/analytics, /api/production
- ProcessingDashboard.tsx	→	/api/harvest
- PoultryCareDashboard.tsx	→	/api/production, /api/medication

Analytics & Dashboard

Frontend:		Backend:
- Dashboard.tsx	→	/api/analytics/dashboard
- Analytics.tsx	→	/api/analytics
- ProductionAnalytics.tsx	→	/api/analytics
- InventoryAnalytics.tsx	→	/api/analytics

System Administration

Frontend:		Backend:
- SystemLogs.tsx	→	/api/system-logs
- ApiKeys.tsx	→	/api/api-keys
- NotificationConfigs.tsx	→	/api/notification-configs
- SystemConfigurationDashboard.tsx	→	/api/configuration
- Settings.tsx	→	/api/configuration

Data Management

Frontend:		Backend:
- DataManagementDashboard.tsx	→	/api/data-management
- Backup configs	→	/api/data-management/backup-configs
- Export configs	→	/api/data-management/export-configs
- Validation rules	→	/api/data-management/validation-rules
- Integrity constraints	→	/api/data-management/integrity-constraints
- Migration plans	→	/api/data-management/migration-plans

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| - Archive policies | → /api/data-management/archive-policies |
| - Operation logs | → /api/data-management/operation-logs |
| - Statistics | → /api/data-management/statistics |

Quality Control

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| Frontend: | Backend: |
| - QualityControlDashboard.tsx | → /api/quality |

Operations Management

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|----------------------------|---------------------------------|
| Frontend: | Backend: |
| - OperationsManagement.tsx | → /api/production, /api/harvest |
| - ProcessingDashboard.tsx | → /api/harvest |
| - Worksheets.tsx | → /api/production |

Employee Management

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|-------------------------|------------------|
| Frontend: | Backend: |
| - EmployeeDashboard.tsx | → /api/employees |
| - EmployeeProfile.tsx | → /api/employees |
| - EmployeeSchedule.tsx | → /api/employees |
| - EmployerDashboard.tsx | → /api/employees |
| - EmployerEmployees.tsx | → /api/employees |

Customer Relationship Management

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|-------------------------|-------------------------|
| Frontend: | Backend: |
| - CustomerDashboard.tsx | → /api/crm, /api/orders |
| - CustomerOrders.tsx | → /api/orders |
| - CustomerCRM.tsx | → /api/crm |

Restaurant Management

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|---------------------------|------------------------------|
| Frontend: | Backend: |
| - RestaurantDashboard.tsx | → /api/orders, /api/products |
| - RestaurantOrders.tsx | → /api/orders |
| - RestaurantInventory.tsx | → /api/inventory |

API Service Integration

The frontend uses `chicken-processing-frontend/src/services/api.ts` to communicate with the backend. This service provides methods for all API calls.

Example API Calls

```
// Authentication
api.post('/auth/login', credentials)
api.post('/auth/register', userData)

// Users
api.get('/users')
api.get('/users/:id')
api.post('/users', userData)
api.put('/users/:id', userData)
api.delete('/users/:id')

// Products
api.get('/products')
api.get('/products/:id')
api.post('/products', productData)
api.put('/products/:id', productData)
api.delete('/products/:id')

// Orders
api.get('/orders')
api.get('/orders/:id')
api.post('/orders', orderData)
api.put('/orders/:id', orderData)
api.put('/orders/:id/cancel', reason)

// Analytics
api.get('/analytics/dashboard')
api.get('/sales', { startDate, endDate })
api.get('/financial', { startDate, endDate })

// Data Management
api.get('/data-management/backup-configs')
api.post('/data-management/backup-configs', config)
api.get('/data-management/statistics')
```

Missing Routes (No Backend Support Yet)

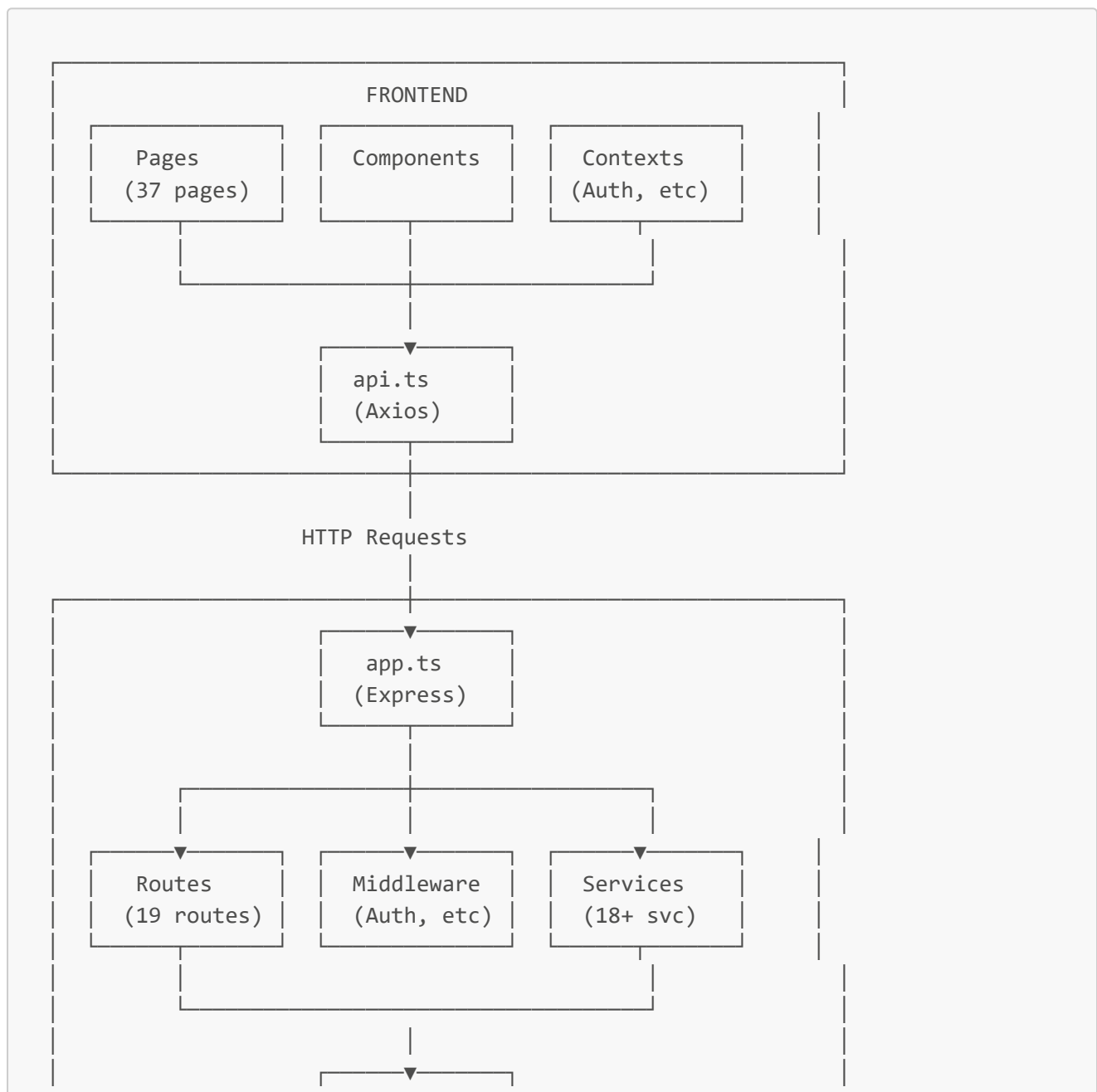
Based on the frontend pages, the following routes might need additional backend support:

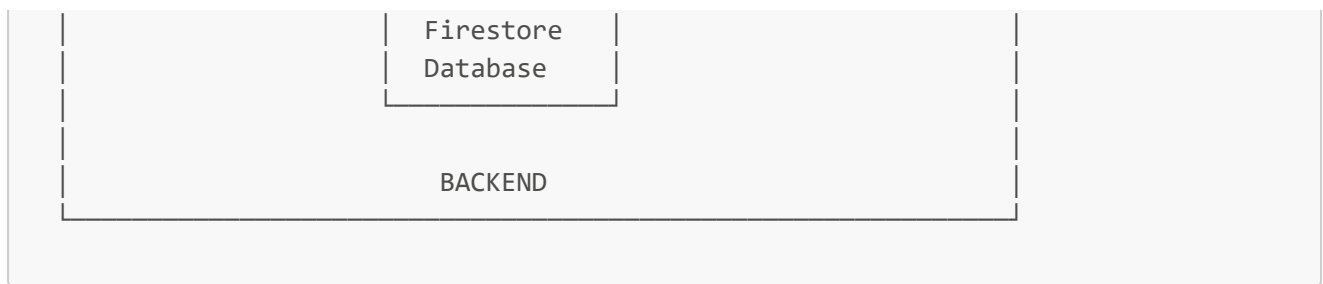
Frontend Page	Potential Missing Route	Notes
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Frontend Page	Potential Missing Route	Notes
Worksheets.tsx	/api/worksheets	Currently uses /api/production
PoultryCareDashboard.tsx	/api/poultry-care	Currently uses /api/production and /api/medication
OperationsManagement.tsx	/api/operations	Currently uses /api/production and /api/harvest

Recommendation: These pages are currently working by combining multiple backend routes, which is a valid approach. Creating dedicated routes is optional and depends on whether you want to simplify the frontend API calls.

Frontend-Backend Communication Flow





Summary

Total Frontend Pages: 37

- **Pages with backend integration:** 32
- **Pages without backend (static/demo):** 5
 - `Home.tsx`
 - `About.tsx`
 - `NotFound.tsx`
 - `ModelBindingDemo.tsx`
 - `Shop.tsx` (uses existing routes)

Total Backend Routes: 19

- **Routes with frontend integration:** 19
- **Routes without frontend:** 0

Coverage: 100%

All backend routes have corresponding frontend pages, and all frontend pages that require backend data have corresponding routes.

Notes

1. **Authentication:** All protected pages use the `AuthContext` to manage user authentication
2. **Protected Routes:** Admin-only pages use `ProtectedRoute` component with role-based access
3. **API Service:** All API calls go through the centralized `api.ts` service
4. **Error Handling:** Backend errors are handled consistently across all pages
5. **Loading States:** All pages implement loading states while fetching data

Future Considerations

1. **Real-time Updates:** Consider adding WebSocket support for real-time notifications
2. **Offline Support:** Consider adding service workers for offline functionality
3. **Caching:** Consider implementing client-side caching for frequently accessed data
4. **Pagination:** All list endpoints should support pagination (already implemented in backend)
5. **Search:** Consider adding global search functionality across all entities